Mth601

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Question No 1:

Maximize $Z = 3x_1 + 5x_2 + 4x_3$

subject to

$$2x_1 + 3x_2 \le 8$$
$$2x_2 + 5x_3 \le 10$$
$$3x_1 + 2x_2 + 4x_3 \le 15$$

and $x_1, x_2, x_3 \ge 0$

Write the standard *form and starting tableau of the linear program* min *g problem*

Question No 2:

Discuss the feasibility of the initial basic solution to a linear programming problem subject to the following constraint.

Maximize $Z=4x_1+3x_2$

Subject to

$$x_1+x_2 \le 50$$
$$x_1+2x_2 \ge 80$$
$$3x_1+2x_2 \ge 140$$
$$x_1,x_2 \ge 0$$

Question No 3:

Convert the following minimization problem into maximization problem

Minimize $Z=x_1-3x_2+2x_3$

Subject to

$$3x_1-x_2+2x_3 \le 7$$
$$-2x_1+4x_2 \le 12$$
$$-4x_1+3x_2+8x_3 \le 10$$
$$x_1,x_2,x_3 \ge 0$$

Question No 4:

Initial basic feasible solution of linear programming problem is given in the following table

Eq.No	**B.Variable**	**Z**	**x_1**	**x_2**	**x_3**	**S_1**	**S_2**	**S_3**	**RHS**	**Ratio**
0	-	1	-3	-5	-4	0	0	0	0	
1	**S_1**	0	2	3	0	1	0	0	8	8/3
2	**S_2**	0	3	2	4	0	1	0	15	15/2
3	**S_3**	0	0	2	5	0	0	1	10	5

Find the next improved solution using the simplex algorithm.

Question No 5:

Express the objective function of following linear programming problem in terms of non-basic variable and hence evaluate the initial basic feasible solution.

Minimize $Z=3x+2y+7z$

Subject to

$-x+y=10$

$2x-y+z \geq 10$

$x,y,z \geq 0$

Solution No 1:

Maximize

$$Z = 3x_1 + 5x_2 + 4x_3 + 0.S_1 + 0.S_2 + 0.S_3$$

Subject to

$$2x_1 + 3x_2 + 0.x_3 + S_1 = 8$$
$$0.x_1 + 2x_2 + 5x_3 + S_2 = 10$$
$$3x_1 + 2x_2 + 4x_3 + S_3 = 15$$
$$x_1, x_2, x_3, S_1, S_2, S_3 \geq 0$$

Tabular Form:

Basic Variable	**Z**	**x_1**	**x_2**	**x_3,**	**S_1**	**S_2**	**S_3**	**RHS**
-	1	-3	-5	-4	0	0	0	0
S_1	0	2	3	0	1	0	0	8
S_2	0	0	2	5	0	1	0	10
S_3	0	3	2	4	0	0	1	15

Solution No 2:

$$x_1 + x_2 + S_1 = 50$$
$$x_1 + 2x_2 - S_2 + A_1 = 80$$
$$3x_1 + 2x_2 - S_3 + A_2 = 140$$
$$x_1, x_2, S_1, S_2, S_3, A_1, A_2 \geq 0$$

Putting decision variable and slack equals to zero

$$x_1 = x_2 = S_2 = S_3 = 0$$

S_1=50

A_1=80

A_2=140

Initial Basic solution is feasible.

Solution No 3:

Maximize $Z = -x_1 + 3x_2 - 2x_3$

Subject to

$$3x_1 - x_2 + 2x_3 \le 7$$
$$-2x_1 + 4x_2 \le 12$$
$$-4x_1 + 3x_2 + 8x_3 \le 10$$
$$x_1, x_2, x_3 \ge 0$$

Solution No 4:

Basic Variable	Z	x_1	x_2	x_3,	S_1	S_2	S_3	RHS
-	1	1/3	0	-4	5/3	0	0	40/3
x_2	0	2/3	1	0	1/3	0	0	8/3
S_2	0	-4/3	0	5	-2/3	1	0	14/3
S_3	0	5/3	0	4	-2/3	0	1	29/3

Solution No 5:

Minimize $Z = 3x + 2y + 7z + MA_1 + MA_2$

subject to

$$-x + y + A_1 = 10$$
$$2x - y + z - S_1 + A_2 = 10$$
$$x, y, z, S_1, A_1, A_2 = 10$$

$A_1 = 10 + x - y$

$A_2 = 10 - 2x + y - z + S_1$

$Z = 3x + 2y + 7z + M(10 + x - y) + M(10 - 2x + y - z + S_1)$

After simplification we get

$Z = (3 - M)x + 2y + z(7 - M) + MS_1 + 20M$

Setting $x = y = S_1 = 0$

$A_1 = 10$

$A_2 = 10$

$Z_{\min} = 20M$

Question No 1:

Discuss differences between maximization and minimization linear programming problems.

Question No 2:

Discuss the degeneracy of the initial basic feasible solution of following linear programming problem.

$$\textit{Maximize} \quad Z = x_1 + x_2 + x_3$$

Subject to

$$x_1 + x_2 \le 8$$
$$-x_2 + x_3 \le 0$$
$$x_1, x_2, x_3 \ge 0$$

Question No 3:

Transform the following linear programming problem into maximization model .Also discuss the degeneracy of the initial feasible solution.

$$\textit{Minimize} \quad Z = -2x_1 + 3x_2$$

Subject to

$$-x_1 + 3x_2 \ge -1$$
$$2x_1 - x_2 = 0$$
$$x_1, x_2 \ge 0$$

Solution No 1:

Maximization:

1) Objective function is of maximization type.

2) Inequalities are of $\leq$ type.

3) Slack variables are added to convert inequality into an equation.

Minimization:

1) Objective function is of minimization type.

2) Inequalities are of $\geq$ type.

3) Slack variables are subtracted and artificial variables are added to convert inequalities into an equation.

Solution No 2:

Maximize $Z = x_1 + x_2 + x_3$

subject to

$$x_1 + x_2 + S_1 = 8$$
$$-x_2 + x_3 + S_2 = 0$$
$$x_1, x_2, x_3, S_1, S_2 \geq 0$$

Setting $x_1 = x_2 = x_3 = 0$

$S_1 = 8$

$S_2 = 0$

One of the basic variable *is zero hence the given solution is* degenearte.

Solution No 3:

Transformed objective function is

Maximize $Z = 2x_1 - 3x_2$

subject to

$$x_1 - 3x_2 \le 1$$
$$2x_1 - x_2 = 0$$
$$x_1, x_2 \ge 0$$

Including artificial var*iable and slack* var*iable*

$$x_1 - 3x_2 + S_1 = 1$$
$$2x_1 - x_2 + A_1 = 0 \Rightarrow A_1 = y - 2x$$
$$x_1, x_2, S_1, A_1 \ge 0$$

Maximize $Z = 2x_1 - 3x_2 - MA$

$$= 2x_1 - 3x_2 - M(x_2 - 2x_1)$$
$$= 2x_1 - 3x_2 - Mx_2 + 2Mx_1$$
$$= 2(1+M)x_1 - (3+M)x_2$$

Setting $x_1 = x_2 = 0$

$S_1 = 1$

$A_1 = 0$

One of the basic var*iable is zero hence the given solution is* deg*enearte.*

Question No 1:

Define the two phase method and Big M method and explain why we prefer two phase method over Big M Method.

Question No 2:

Transform the following linear programming problem in the first Phase by following the two phase method

$$Minimize. Z = x_1 + x_2$$

$$Subject\ to:$$

$$2x_1 + x_2 \geq 4$$

$$x_1 + 7x_2 \geq 7$$

$$x_1, x_2 \geq 0$$

Question No 3:

By using the two phase method express the objective function of the linear programming problem in terms of non-basic variables in first phase.

$$Minimize \quad Z = 4x + y$$

$$Subject\ to:$$

$$3x+y=3$$

$$4x + 3y \geq 6$$

$$x + 2y \leq 4$$

$$x, y \geq 0$$

Question No 4:

Write standard form of first phase of following linear programming problem.

$$Minimize \quad Z = 4x_1 + x_2 + x_3$$

$$Subject\ to:$$

$$2x_1 + x_2 + 2x_3 = 4$$

$$3x_1 + 3x_2 + x_3 = 3$$

$$x_1, x_2, x_3 \geq 0$$

Solution No 2:

Minimize $Z = A_1 + A_2$

Subject to

$$2x_1 + x_2 + A_1 = 4$$
$$x_1 + 7x_2 + A_2 = 7$$
$$x_1, x_2, A_1, A_2 \geq 0$$

Solution No 3:

Minimize $Z = A_1 + A_2$

Subject to

$$3x + y + A_1 = 3$$
$$4x + 3y - S_1 + A_2 = 6$$
$$x + 2y + S_2 = 4$$
$$x, y, A_1, A_2, S_1, S_2 \geq 0$$

where

$$A_1 = 3 - 3x - y$$
$$A_2 = 6 - 4x - 3y + S_1$$

Since the objective function is exp*ressed in terms of basic* var*iables so to transform these* int*o non basic Pu*
A_1*and* A_2 *from constra* int*s* int*o objective function.*

$Z = 3 - 3x - y + 6 - 4x - 3y + S_1$

$Z = 9 - 7x - 4y + S_1$

Solution No 4:

Minimize $Z = A_1 + A_2$

subject to

$$2x_1 + x_2 + 2x_3 + A_1 = 4$$
$$3x_1 + 3x_2 + x_3 + A_2 = 3$$
$$x_1, x_2, x_3, A_1, A_2 \geq 0$$

Question No 1:

Write the given linear programming problem in standard form and discuss the nature of initial basic feasible solution

$$\textit{Maximize} \quad Z = 2x_1 + 3x_2 + 10x_3$$

subject to

$$x_1 + 2x_3 = 0$$
$$x_2 + x_3 = 1$$
$$x_1, x_2, x_3 \geq 0$$

Question No 2:

Explain degeneracy in terms of graphical method.

Question No 3:

B.variable	**x_1**	**x_2**	**S_1**	**S_2**	**S_3**	**Solution**
Z	0	-5/4	0	3/4	0	6
S_1	0	2	1	-1	0	4
x_1	1	1/4	0	1/4	0	2
S_3	0	-2	0	-1	1	0

Operate the next iteration and discuss the solution.

Question No 4:

$$\textit{Maximize} \quad Z = 2x_1 + 5x_2$$

subject to

$$x_1 \leq 40$$
$$x_2 \leq 30$$
$$x_1 + x_2 \leq 30$$
$$x_1, x_2 \geq 0$$

Operate the next iteration and discuss the solution.

Solution No 1:

Maximize $Z = 2x_1 + 3x_2 + 10x_3$

subject to

$$x_1 + 2x_3 + A_1 = 0$$
$$x_2 + x_3 + A_2 = 1$$
$$x_1, x_2, x_3, \mathrm{A}_1, \mathrm{A}_2 \geq 0$$

Setting $x_1 = x_2 = x_3 = 0$

$A_1 = 0$

$A_2 = 1$

One of the basic variable is zero in the initial stage,so it is degenerate basic solution.

Solution No 3:

B.variable	**x_1**	**x_2**	**S_1**	**S_2**	**S_3**	**Solution**
Z	0	0	5/8	1/8	0	17/2
x_2	0	1	1/2	-1/2	0	2
x_1	1	0	-1/8	3/8	0	3/2
S_3	0	0	1	-2	1	4

Solution is temporarily degenerate

Solution No 4:

Tie between S_2 and S_3

B.variable	**Z**	**x_1**	**x_2**	**S_1**	**S_2**	**S_3**	**RHS**
	1	-2	0	0	5	0	150
S_1	0	1	0	1	0	0	40
x_2	0	0	1	0	1	0	30
S_3	0	1	0	0	-1	1	0

Second iteration:

B.variable	**Z**	**x_1**	**x_2**	**S_1**	**S_2**	**S_3**	**RHS**
	1	0	0	0	3	2	150
S_1	0	0	0	1	1	-1	40
x_2	0	0	1	0	1	0	30
x_1	0	1	0	0	-1	1	0

Basic variable x is equal to zero .This implies degenerate optimal solution.

Question No 1:

How do we come to know that the given Linear Programming problem has multiple optimal solutions by solving it by simplex method. Also discuss it for graphical method.

Question No 2:

Discuss the following

a) Unbounded solution of linear programming Problem

b) Infeasible solution of linear programming problem

Question No 3:

Put the following linear programming problem of unrestricted decision variables in standard form.

$$\begin{aligned} &\textit{Maximize} && Z = 2x_1 + 3x_2 \\ &\textit{subject to} && -x_1 + 2x_2 \le 4 \\ & && x_1 + x_2 \le 6 \\ & && x_1 + 3x_2 \le 9 \\ & && x_1, x_2 \ \textit{unrestricted} \end{aligned}$$

Question No 4:

$$\begin{aligned} &\textit{Maximize} && Z = 3x_1 + 4x_2 \\ &\textit{subject to} && 2x_1 + x_2 \le 4 \\ & && x_1 + 2x_2 \ge 12 \\ & && x_1, x_2 \ \ge 0 \end{aligned}$$

Solve the above linear programming problem and discuss the type of solution

Question No 5:

Given linear programming problem is of maximization. Operate the next iteration and discuss the type of solution.

B.variable	**x_1**	**x_2**	**x_3**	**x_4**	**x_5**	**A**	**Solution**
Z	-20	-10	-1	0	0	0	0
x_4	3	-3	5	1	0	0	50
x_5	1	0	1	0	1	0	10
x_6	1	-1	4	0	0	1	20

Solution No 3:

$x_1 = x_1' - x_1''$

$x_2 = x_2' - x_2''$

$Z = 2(x_1' - x_1'') + 3(x_2' - x_2'') + 0.S_1 + 0.S_2 + 0.S_3$

s.t

$-x_1' + x_1'' + 2x_2' - 2x_2'' + S_1 = 4$

$x_1' - x_1'' + x_2' - x_2'' + S_2 = 6$

$x_1' - x_1'' + 3x_2' - 3x_2'' + S_3 = 9$

Taking $x_1' = x_1'' = x_2' = x_2'' = 0$

$S_1 = 4$

$S_2 = 6$

$S_3 = 9$

Solution No 4:

B.variable	**Z**	**x_1**	**x_2**	**S_1**	**S_2**	**A**	**RHS**
	1	-3-M	-4-2M	0	M	0	-12M
S_1	0	2	1	1	0	0	4
A	0	1	2	0	-1	1	12

Second iteration

B.variable	**Z**	**x_1**	**x_2**	**S_1**	**S_2**	**A**	**RHS**
	1	M+1	0	4+2M	M	0	-4M+16
x_2	0	2	1	1	0	0	4
A	0	-3	0	-2	-1	1	4

We cannot proceed further with maximization as there is no negative coefficient in the objective row but the final solution has the artificial variable in the basis with the value equal to 4.Hence the problem has no feasible optimal solution.

Solution No 5:

Basic Variable	**X_1**	**X_2**	**X_3**	**X_4**	**X_5**	**R**	**Solution**
Z	**0**	**-10**	**19**	**0**	**20**	**0**	**200**
X_4	0	-3	2	1	-3	0	20
X_1	1	0	1	0	1	0	10
X_6	0	-1	3	0	-1	1	10

Because the elements of pivot column are zero and negative, so the ratio does not have a proper meaning. So the solution is unbounded.

Question No 1:

Write down the dual of the following linear programming problem.

a)

$$\begin{aligned} &\textit{Maximize} && Z = w_1 + 2w_2 + w_3 \\ &\textit{subject to} \\ & && 2w_1 + w_2 - w_3 \le 2 \\ & && -2w_1 + w_2 - 5w_3 \ge -4 \\ & && 4w_1 + w_2 + w_3 \le 6 \\ & && w_1, w_2, w_3 \ge 0 \end{aligned}$$

b)

$$\begin{aligned} &\textit{Minimize} && Z = 3x_1 + x_2 \\ &\textit{subject to} \\ & && 2x_1 + 3x_2 \ge 2 \\ & && x_1 + x_2 \ge 1 \\ & && x_1, x_2 \ge 0 \end{aligned}$$

c)

$$\begin{aligned} &\textit{Minimize} && Z = 2x_2 + 5x_3 \\ &\textit{subject to} \\ & && x_1 + x_2 \ge 2 \\ & && 2x_1 + x_2 + 6x_3 \le 6 \\ & && x_1 - x_2 + 3x_3 = 4 \\ & && x_1, x_2, x_3 \ge 0 \end{aligned}$$

d)

$$\begin{aligned} &\textit{Minimize} && Z = 4y_1 + 3y_2 + 5y_3 \\ &\textit{subject to} \\ & && y_1 + y_2 - y_4 \ge 10 \\ & && -y_1 - y_2 + y_4 \le -10 \\ & && y_1 - y_5 = 0 \\ & && y_2 + y_6 = 0 \\ & && y_3 - y_7 = 0 \\ & && y_1, y_2, y_3, y_4, y_5, y_6, y_7 \ge 0 \end{aligned}$$

Question No 2:

What are the advantages of using duality theory? Explain with the help of an example.

Question No 1:

Determine whether the following Transportation model has initial feasible solution?

	A	**B**	**C**	Supply
1	$\mathbf{x}_{11}$	$\mathbf{x}_{12}$	$\mathbf{x}_{13}$	5
2	$\mathbf{x}_{21}$	$\mathbf{x}_{22}$	$\mathbf{x}_{23}$	4
3	$\mathbf{x}_{31}$	$\mathbf{x}_{32}$	$\mathbf{x}_{33}$	9
Demand	4	6	8	

Question No 2:

Determine whether the following Transportation model has initial feasible solution?

Plant	$\boldsymbol{W}_1$	$\boldsymbol{W}_2$	$\boldsymbol{W}_3$	$\boldsymbol{W}_4$	*Supply*
$\boldsymbol{F}_1$	11	20	7	8	50
$\boldsymbol{F}_2$	21	16	10	12	40
$\boldsymbol{F}_3$	8	12	18	9	70
Demand	30	25	35	40	

Question No 3:

Balance the following Transportation Problem and hence construct the modified table.

	1	2	3	4	Supply
A	x_{11}	x_{12}	x_{13}	x_{14}	7
B	x_{21}	x_{22}	x_{23}	x_{24}	7
C	x_{31}	x_{32}	x_{33}	x_{34}	5
Demand	5	5	8	4	

Question No 4:

State the transportation problem in general terms and explain how it is tested whether a basic feasible solution of a transportation problem is optimal or not?

Solution No 1:

Total demand = 4 + 6 +8 =18
Total Supply=5+4+9=18
Since total supply is equal to total demand
Given problem has the initial basic feasible solution.

Solution No 2:

Total demand = 30+25+35+40=130
Total Supply=50+40+70=160
Since total supply is not equal to total demand
Given problem has not initial basic feasible solution.

Solution No 3:

Total demand = 5+5+8+4=22
Total Supply=7+7+5=19
Since total demand is greater than total supply the given problem is unbalance.
Therefore introduce a dummy row at a unit transportation cost of zero and having the availability of 3 items. The modified table is

	1	2	3	4	Supply
A	x_{11}	x_{12}	x_{13}	x_{14}	7
B	x_{21}	x_{22}	x_{23}	x_{24}	7
C	x_{31}	x_{32}	x_{33}	x_{34}	5
D	0	0	0	0	3
Demand	5	5	8	4	

Now, Total supply=7+7+5+3=22=5+5+8+4=Total demand

Given problem is balanced and has the initial basic feasible solution

Question No 1:

Find the initial basic feasible solution of a transportation problem using least cost method.

	W_1	W_2	W_3	Supply
F_1	2	7	4	5
F_2	3	3	1	8
F_3	5	4	7	7
F_4	1	6	2	14
Demand	7	9	18	34

Question No 2:

Determine the initial basic feasible solution of the following transportation problem using the north-west corner rule.

	D_1	D_2	D_3	D_4	Supply
P_1	6	4	1	5	14
P_2	8	9	2	7	16
P_3	4	3	6	2	5
Demand	6	10	15	4	35

Question No 3:

Solve the following transportation problem using the Vogel's approximation method.

	1	2	3	Supply
A	2	7	4	5
B	3	3	1	8
C	5	4	7	7
D	1	6	2	14
Demand	7	9	18	34

Question No 4:

Check whether the given initial basic feasible solution is optimal or not?

19	14	23	11(11)
15(6)	16(3)	12	21(4)
30	25(7)	16(12)	39

Question No 5:

	1	**2**	**3**	**4**	**Supply**
A	10	2	20	11	15
B	12	7	9	20	25
C	4	14	16	18	10
Demand	5	15	15	15	

Find the following using Pij = **U*i*** + **V*j*** – **C*ij***

a) P_{14}

b) P_{23}

c) P_{31}

d) P_{32}

Solution No 1:

Examine the cost cell carefully and choose the cell with the lowest cost and allocate there as much as possible.

	W_1	**W_2**	**W_3**	**Supply**
F_1	2	7	4	5
F_2	3	3	1(8)	0
F_3	5	4	7	7
F_4	1	6	2	14
Demand	7	9	10	

Since F_2 is exhausted so eliminate that row

	W_1	**W_2**	**W_3**	**Supply**
F_1	2	7	4	5
F_3	5	4	7	7
F_4	1(7)	6	2	7
Demand	0	9	10	

Since W_1 is exhausted so eliminate that column

	W_2	**W_3**	**Supply**
F_1	7	4	5
F_3	4	7	7
F_4	6	2(7)	0
Demand	9	3	

Since F_4 is exhausted so eliminate that row

	W_2	**W_3**	**Supply**
F_1	7	4	5
F_3	4(7)	7	0
Demand	2	3	

Since F_3 is exhausted so eliminate that row

	W_2	**W_3**	**Supply**
F_1	7(2)	4(3)	5
Demand	2	3	

Solution No 2:

	D_1	**D_2**	**D_3**	**D_4**
P_1	6(6)	4(8)	1	5
P_2	8	9(2)	2(14)	7
P_3	4	3	6(1)	2(4)

Total transportation cost by this method is

=(6*6)+(4*8)+(9*2)+(2*14)+(6*1)+(2*4)

=36+32+18+28+6+8=68+46+14=128

Solution No 3:

	1	**2**	**3**	**Supply**	Penalities
A	2	7	4	5	(2)
B	3	3	1(8)	8	(2)
C	5	4	7	7	(1)
D	1	6	2	14	(1)
Demand	7	9	18	34	
	(1)	(1)	(1)		

	1	**2**	**3**	**Supply**	Penalities
A	2	7	4	5	(2)
C	5	4	7	7	(1)
D	1	6	2(10)	14	(1)
Demand	7	9	18		
	(1)	(2)	(2)		

	1	2	3	Supply	Penalities
A	2	7	4	5	(2)
C	5	4	7	7	(1)
D	1	6	2(10)	14	(1)
Demand	7	9	10		
	(1)	(2)	(2)		

	1	2	Supply	Penalities
A	2	7	5	(5)
C	5	4	7	(1)
D	1(4)	6	4	(5)
Demand	7	9		
	(1)	(2)		

	1	2	Supply	Penalities
A	2(3)	7(7)	5	(5)
C	5	4(7)	7	(1)
Demand	3	9		
	(3)	(3)		

Thus we get the required feasible solution as shown in the following table

	1	2	3	Supply
A	2(3)	7(2)	4	5
B	3	3	1(8)	8
C	5	4(7)	7	7
D	1(4)	6	2(10)	14
Demand	7	9	18	34

The total transportation cost to this feasible solution

=6+14+8+28+4+20=80

Solution No 4:

Matrix of cost of allotted cells

			11
15	16		21
	25	16	39

To determine u_i's and v_j's we arbitrarily assign one of the multiplier say u_2 equal to zero

			11	u_1
15	16		21	u_2
	25	16	39	u_3
v_1	v_2	v_3	v_4	

$u_1+ v_4=11$ ⇒ $u_1=11- v_4$ ⇒ $u_1=11-21=-10$ ⇒ $u_1=-10$

$u_2+ v_1=15$ ⇒ $v_1=15- u_2$ ⇒ $v_1=15$

$u_2+ v_2=16$ ⇒ $v_2=16- u_2$ ⇒ $v_2=16$

$u_2+ v_4=21$ ⇒ $v_4=21- u_2$ ⇒ $v_4=21$

$u_3+ v_2=25$ ⇒ $u_3=25- v_2$ ⇒ $u_3=25-16=9$ ⇒ $u_3=9$

$u_3+ v_3=16$ ⇒ $v_3=16- u_3$ ⇒ $v_3=16- 9=7$ ⇒ $v_3=7$

Cost matrix of unoccupied cells

19	14	23	
		12	
30			39

u_i and v_j for unoccupied cells

$u_1+ v_1=-10+15=5$

$u_1+ v_2=-10+16=6$

$u_1+ v_3=-10+7=-3$

$u_2+ v_3=0+7=7$

$u_3+ v_1=9+15=24$

$u_3+ v_4=9+21=30$

Calculate $c_{ij}-(u_i+v_j)$ for each of unoccupied cells

14	8	26	
		5	
6			9

Since all the values are positive therefore the solution is optimal.

Solution No 5:

Do it by yourself.